ARHAM ZAHID KHAN

Software & AI Engineer | Full-Stack & LLM Agent Architecture

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PROFESSIONAL SUMMARY

Final-year BS Computer Science student and Software & AI Engineer with strong expertise in Object-Oriented Programming, Full-Stack Web Development, Generative AI, and LLM Agent Architecture. Co-developing CareerSyncAI, a dual-sided AI recruitment platform integrating FastAPI, PostgreSQL, Next.js, and multi-agent workflows (LangChain, CrewAI, Gemini/Claude APIs). Proficient in building production-ready REST APIs, structured prompt contracts, evaluation suites, and scalable web applications. Seeking an AI Engineer role to build high-performance, enterprise-grade AI systems.

TECHNICAL SKILLS

- **Programming Languages:** Python, TypeScript, JavaScript (ES6+), C++, SQL
- **AI & LLM Systems:** Multi-Agent Workflows (LangChain, CrewAI), Gemini & Claude APIs, Prompt Engineering & Contracts, Structured Output Schemas (Pydantic), Ground-Truth Eval Suites
- **Backend & APIs:** FastAPI, RESTful APIs, JWT Authentication, Swagger/OpenAPI, Firebase (Auth, Firestore)
- **Frontend Development:** Next.js, React.js, Tailwind CSS, HTML5, CSS3, Responsive UI Development
- **Databases & ORM:** PostgreSQL, MySQL, Schema Design, Alembic Migrations, SQL Query Optimization (B-Tree Indexing)
- **DevOps & Tools:** Git, GitHub (Branching Workflows), VS Code, Linux (Ubuntu), Docker, Postman
- **Software Engineering:** Object-Oriented Programming (OOP), Data Structures & Algorithms, System Architecture, UI/UX (Figma, Adobe Creative Suite)

WORK EXPERIENCE

AI Automation & Prompt Engineer Intern | Developers Hub, Lahore **Aug 2026 – Oct 2026**

- Engineered prompt validation contracts, system prompts, and custom Claude skills defining task contexts, JSON schemas, and fallback rules.
- Built evaluation suites covering edge cases, malformed inputs, and prompt-injection risks with stage-wise failure logging.
- Engineered interactive clarification loops allowing AI agents to handle incomplete user inputs effectively.
- Documented automated workflows and test failure logs for team reusability and production deployment maintainability.

Graphic Design Intern | NIAIS (National Initiative for AI & Security), Lahore **Dec 2022 – Jan 2023**

- Designed digital banners, visual assets, and marketing collateral for national AI awareness campaigns under tight deadlines.

KEY PROJECTS

CareerSyncAI – Dual-Sided AI Recruitment Platform | Final Year Project **2026 – Present**

- Tech Stack: Python, FastAPI, Next.js, TypeScript, PostgreSQL, LangChain, CrewAI, Gemini / Claude API, Tailwind CSS
- Engineering a dual-sided hiring platform integrating AI multi-agent workflows for automated resume parsing, CV tailoring, and candidate screening.
- Developing REST APIs in FastAPI with JWT auth, Pydantic structured schemas, and PostgreSQL database migrations (Alembic).
- Building responsive frontend interfaces in Next.js & Tailwind CSS featuring candidate and recruiter dashboards with Kanban hiring pipelines.

House Price Prediction Model | Machine Learning Project **2026 – Present**

- Tech Stack: Python, Pandas, NumPy, Scikit-Learn, Matplotlib, Jupyter Notebook
- Built a machine learning regression pipeline to predict real estate property prices based on structural and spatial features.
- Executed data preprocessing, feature engineering, missing value imputation, and correlation analysis to optimize model accuracy (RMSE).

Order Tracker App | Full-Stack Web App **2026**

- Tech Stack: React.js, Firebase Auth, Cloud Firestore, Tailwind CSS
- Built a real-time order tracking web application featuring Google Sign-In and secure user order data isolation.

- Optimized Firestore real-time listeners inside custom React hooks to prevent re-render quota over-consumption.

Smart Property Rental Management System | Relational DBMS **2025**

- Tech Stack: SQL, MySQL, Relational Database Design
- Designed a normalized 3NF relational database schema managing lease records, tenant ledgers, and maintenance logs.
- Optimized query execution times on large test datasets using B-Tree indexing strategies and optimized views.

Brick Breaker Game | C++ & Systems Project **2024**

- Tech Stack: C++, OOP Architecture, Physics Engine
- Developed a 2D arcade game utilizing dynamic memory management, OOP design patterns, and continuous collision detection math.

EDUCATION

BS Computer Science | University of Lahore, Pakistan **Sep 2023 – Present**

- Final Year (7th Semester). Core Focus: OOP, Data Structures & Algorithms, DBMS, Web Development, Software Engineering.

FSc Pre-Engineering | Govt. Shalimar College, Lahore **2020 – 2023**

Matriculation (Computer Science) | Govt. High School Baghbanpura, Lahore **2018 – 2020**

CERTIFICATIONS

- AI Automation & Agent Architecture Masterclass (System Prompts, Claude Skills, MCP, Eval Suites)
- Introduction to Front-End Development – Coursera
- Programming with JavaScript – Coursera
- Foundations of User Experience (UX) Design – Google / Coursera
- Build Wireframes and Low-Fidelity Prototypes – Google / Coursera